

# **SHORT-TERM FERRY TICKET DEMAND FORECASTING & PREDICTIVE DECISION SUPPORT SYSTEM**

*A Machine Learning Approach to Short-Horizon Demand Forecasting  
for Toronto Island Park Ferry Operations*

## **Research Paper**

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Dataset: Toronto Island Ferry Ticket Sales & Redemptions, May 2015 – December  
2025

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# Abstract

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Toronto Island Park's ferry service currently relies on reactive, after-the-fact analysis of ticketing data, leaving operators without advance warning of short-term demand surges. This project develops and evaluates a short-term (15-minute to 2-hour horizon) ferry ticket demand forecasting system using ten years of 15-minute-interval sales and redemption data (May 2015 – December 2025; 261,538 raw records). After identifying and resolving significant data quality issues — including implicit zero-activity gaps covering 29.9% of the theoretical time grid and a small number of extreme outlier intervals — a suite of forecasting approaches was developed and compared, ranging from naive persistence baselines through linear regression to gradient-boosted decision trees.

The Gradient Boosting model consistently outperformed all baseline approaches, and the performance gap widened substantially at longer forecast horizons: naive and moving-average approaches degraded by 77–104% in average error between the 15-minute and 2-hour horizons, while Gradient Boosting degraded by only 22%, indicating it captures durable demand patterns rather than simple recent-value persistence. Quantile regression was used to produce well-calibrated 80% prediction intervals, shown to adapt meaningfully to operating conditions — averaging under 10 tickets wide on quiet winter days and over 130 tickets wide on peak summer weekends. These findings were operationalized into an interactive dashboard enabling ferry operations staff to view forecasts, compare models, and inspect uncertainty ranges for any date, time, and horizon within the evaluation window. This paper documents the data preparation methodology, modeling approach, results, and operational recommendations for deployment.

## 1. Introduction

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### 1.1 Background

Toronto Island Park is served by a ferry system that experiences highly variable demand across the day, the week, and the season, driven by weather, park programming, and seasonal visitation patterns. Historically, operational decisions — how many ferries to run, how much staff to schedule, when to prepare for crowding — have been informed by retrospective reporting rather than forward-looking prediction.

### 1.2 Problem Statement

Despite over a decade of high-frequency ticketing data being collected, no system currently exists to convert this historical record into short-term operational forecasts. This leaves ferry operations in a reactive posture: congestion and demand spikes are identified only after they occur, rather

than anticipated with enough lead time to adjust staffing or service frequency.

### 1.3 Objectives

Primary objectives:

- Forecast short-term ferry ticket sales and redemptions
- Predict demand for upcoming 15-minute to 2-hour windows
- Compare statistical and machine-learning forecasting approaches

Secondary objectives:

- Quantify prediction uncertainty
- Support proactive operational planning
- Demonstrate a deployable forecasting interface

## 2. Data

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### 2.1 Source and Structure

The dataset consists of 261,538 raw records spanning May 1, 2015 to December 21, 2025 (approximately 10.6 years), with the following fields:

| Column           | Description                      |
|------------------|----------------------------------|
| _id              | Unique row identifier            |
| Timestamp        | 15-minute interval end time      |
| Sales Count      | Tickets sold in the interval     |
| Redemption Count | Tickets redeemed in the interval |

### 2.2 Initial Data Quality Assessment

Two issues surfaced during initial inspection:

- A small number of extreme values, headlined by a single interval (August 15, 2023, 20:15) recording 7,229 sales and 7,216 redemptions — roughly ten times the 99.9th percentile of the rest of the distribution, consistent with a data correction or batch-upload artifact rather than organic demand.
- Substantial apparent gaps in the timeline: only 261,538 of the 373,189 theoretically possible 15-minute slots between the first and last record contained a row — a 29.9% gap rate.

## 3. Data Cleaning & Preprocessing

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### 3.1 Resolving Missing Intervals

A key diagnostic check was performed before assuming the missing intervals represented a data collection failure: across all 261,538 raw records, not a single row recorded zero sales and zero redemptions simultaneously. This indicates the source system only logs an interval when at least one transaction occurs — meaning every “missing” 15-minute slot in fact represents an interval of literal zero ticket activity, not lost data.

Accordingly, the full 15-minute timeline was reconstructed and every missing slot was filled with (0, 0) sales/redemptions, producing a complete, gap-free dataset of 373,189 rows.

### 3.2 Outlier Treatment

Extreme values were capped at the 99.9th percentile of each series (707.9 tickets for Sales Count, 736.5 for Redemption Count) to prevent a handful of anomalous intervals from disproportionately influencing model training and error metrics, particularly RMSE. This affected 409 rows (0.11% of the dataset). Original, uncapped values were preserved in parallel columns, and each affected row was flagged, to keep the transformation fully auditable.

### 3.3 Cleaning Summary

| Metric                               | Value                      |
|--------------------------------------|----------------------------|
| Raw records                          | 261,538                    |
| Date range                           | May 1, 2015 - Dec 21, 2025 |
| Complete 15-min grid (post-fill)     | 373,189 rows               |
| Missing slots filled (implicit zero) | 111,651 (29.9%)            |
| Outlier-capped rows                  | 409 (0.11%)                |
| Sales Count cap (99.9th pct.)        | 707.9                      |
| Redemption Count cap (99.9th pct.)   | 736.5                      |

## 4. Exploratory Data Analysis

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### 4.1 Intraday Pattern

Aggregate hourly averages show a pronounced single-peaked daily pattern: activity is negligible overnight (the 4:00 a.m. hour averages just 0.22 tickets sold per 15-minute interval), rises through the morning, and peaks at noon

(90.9 tickets sold per interval on average) before declining through the afternoon and evening.

## 4.2 Seasonal Pattern

Ticket volume is highly seasonal. Aggregating total sales by calendar month across all years, August is the highest-volume month (3,407,307 total tickets) and January the lowest (127,928) — a 26.6-fold difference, consistent with Toronto Island Park's role as a warm-weather recreational destination.

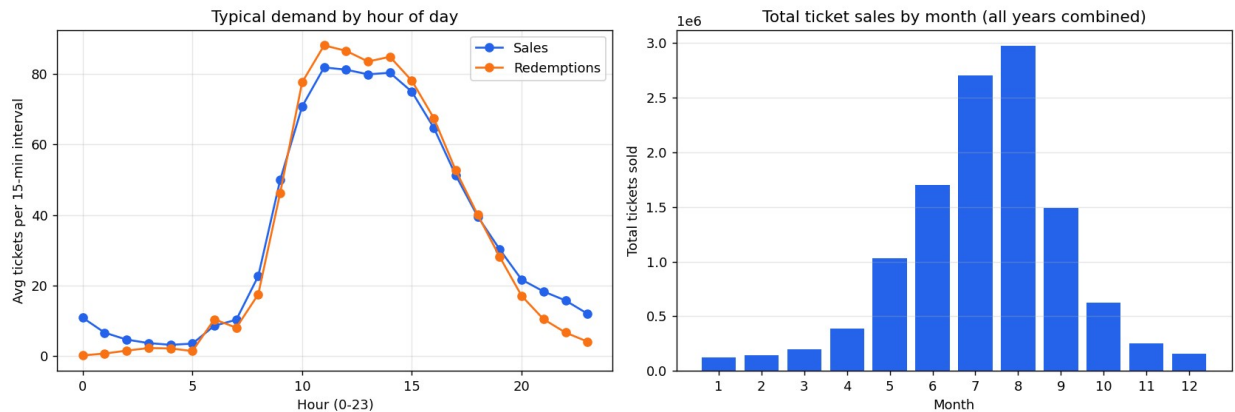


Figure 1. Left: average ticket activity by hour of day. Right: total ticket sales by calendar month, all years combined.

## 5. Feature Engineering

To convert the cleaned time series into a supervised learning problem, the following predictor variables were derived for every interval, using only information available prior to (not including) the interval being predicted, to prevent data leakage:

- Lag features: sales and redemption counts at 15, 30, 60, and 120 minutes prior
- Rolling statistics: mean, standard deviation, and maximum over the preceding hour (four intervals), computed strictly on past values
- Temporal encodings: hour of day, day of week, calendar month, and a weekend indicator

## 6. Methodology

### 6.1 Train/Test Split

A strict time-based split was used, as required for valid time-series evaluation: the final six months of the dataset (June 22, 2025 – December 21, 2025; 17,473 intervals) were held out as a test set, with all preceding data

(355,708+ intervals) used for training. No random shuffling was performed at any stage, ensuring the evaluation reflects genuine forward-looking forecast performance rather than in-sample fit.

## 6.2 Models Evaluated

Four primary approaches were compared at every horizon:

- Naive persistence (forecast = most recently observed value)
- Moving average (forecast = mean of the trailing hour)
- Linear regression on the engineered feature set
- Gradient Boosting Regressor (scikit-learn) on the engineered feature set

A Random Forest Regressor was additionally evaluated at the 15-minute horizon for comparison (Section 7.1). Two model families named in the original project scope — ARIMA/SARIMA and Facebook Prophet — could not be evaluated in the development environment used for this phase, as the required packages could not be installed without internet access. Section 10 outlines a recommended path to include them in a follow-up iteration.

## 6.3 Multi-Horizon Forecasting

Separate models were trained to predict four forecast horizons — 15 minutes, 30 minutes, 1 hour, and 2 hours ahead — each using an identical input feature set but a target value shifted to the corresponding future interval. Careful attention was paid to correctly aligning each horizon's target with the point in time at which the forecast would actually be issued; an initial implementation was found during development to be systematically misaligned by one 15-minute step across all horizons, and was corrected prior to finalizing the results reported in Section 7. This is noted here as it illustrates a common and easily-overlooked failure mode in time-series pipeline construction, and the correction process is documented for reproducibility.

## 6.4 Uncertainty Quantification

For the Gradient Boosting model, quantile regression (10th and 90th percentile loss functions) was used to produce an 80% prediction interval around each point forecast, in addition to the standard mean-based prediction.

# 7. Results

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## 7.1 Model Comparison (15-Minute Horizon)

Table 1 compares all evaluated models at the 15-minute forecast horizon. Random Forest and Gradient Boosting performed comparably at this horizon and both substantially outperformed the baseline approaches; Gradient Boosting was selected as the production model on the strength of its

performance at longer horizons (Section 7.2) and its compatibility with quantile-based uncertainty estimation (Section 7.4).

*Table 1. Model comparison at the 15-minute forecast horizon, evaluated on the held-out test period.*

| Model                 | MAE (tickets) | RMSE  |
|-----------------------|---------------|-------|
| Naive (last value)    | 19.38         | 42.83 |
| Moving Average (1 hr) | 17.90         | 37.57 |
| Linear Regression     | 17.37         | 35.67 |
| Random Forest         | 15.41         | 33.17 |
| Gradient Boosting     | 15.42         | 33.22 |

## 7.2 Horizon-Wise Performance

*Table 2. Mean Absolute Error (MAE) by model and forecast horizon.*

| Horizon | Naive | Moving Avg | Linear Reg | Gradient Boosting |
|---------|-------|------------|------------|-------------------|
| 15 min  | 19.38 | 17.90      | 17.37      | 15.42             |
| 30 min  | 20.81 | 20.00      | 18.86      | 16.07             |
| 1 hour  | 24.18 | 25.07      | 22.25      | 16.77             |
| 2 hours | 34.30 | 36.50      | 30.01      | 18.80             |

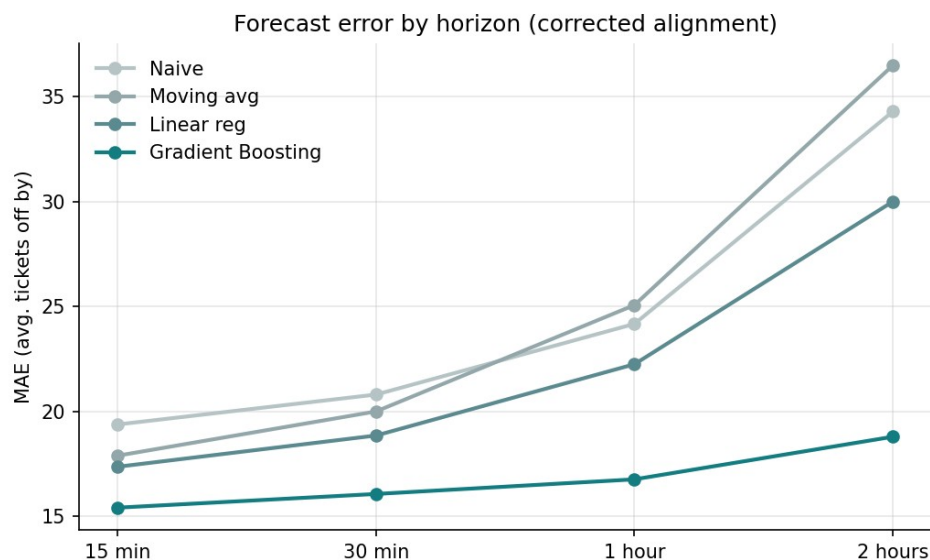


Figure 2. Forecast error (MAE) by horizon, corrected for timestamp alignment (Section 6.3).

Naive and moving-average approaches degrade sharply as the horizon lengthens (Naive: 19.38  $\rightarrow$  34.30 MAE, a 77% increase; Moving Average: 17.90  $\rightarrow$  36.50, a 104% increase), while Gradient Boosting degrades only modestly (15.42  $\rightarrow$  18.80, a 22% increase). This indicates the engineered features — particularly recent-trend and temporal signals — retain predictive value well beyond the immediate next interval, whereas simpler methods have no mechanism for anticipating change over longer horizons.

### 7.3 Feature Importance

Examining Gradient Boosting / Random Forest feature importances at the 15-minute horizon, the rolling mean of the preceding hour accounts for approximately 90% of total feature importance, with the most recent lag value, hour-of-day, and rolling volatility (standard deviation) contributing most of the remainder. Calendar features (day of week, month, weekend indicator) contribute comparatively little at this short horizon, though their relative contribution would be expected to grow for longer-horizon or daily-level forecasts.

### 7.4 Prediction Interval Calibration

Table 3. Prediction interval coverage and width by horizon (Gradient Boosting, 80% nominal interval).

| Horizon | Target Coverage | Actual Coverage | Avg. Band Width |
|---------|-----------------|-----------------|-----------------|
| 15 min  | 80%             | 86.6%           | 47.0 tickets    |
| 30 min  | 80%             | 86.9%           | 49.7 tickets    |
| 1 hour  | 80%             | 87.3%           | 54.7 tickets    |
| 2 hours | 80%             | 87.7%           | 65.4 tickets    |

Across all four horizons, the 80% prediction interval achieved empirical coverage between 86.6% and 87.7% — modestly conservative rather than under-confident, which is the safer direction of error for an operational planning tool. Interval width scaled meaningfully with real operating conditions: on the quietest test-period days the average band was under 10 tickets wide, while on the busiest summer weekend dates it exceeded 130 tickets — more than a tenfold difference (Figure 3).

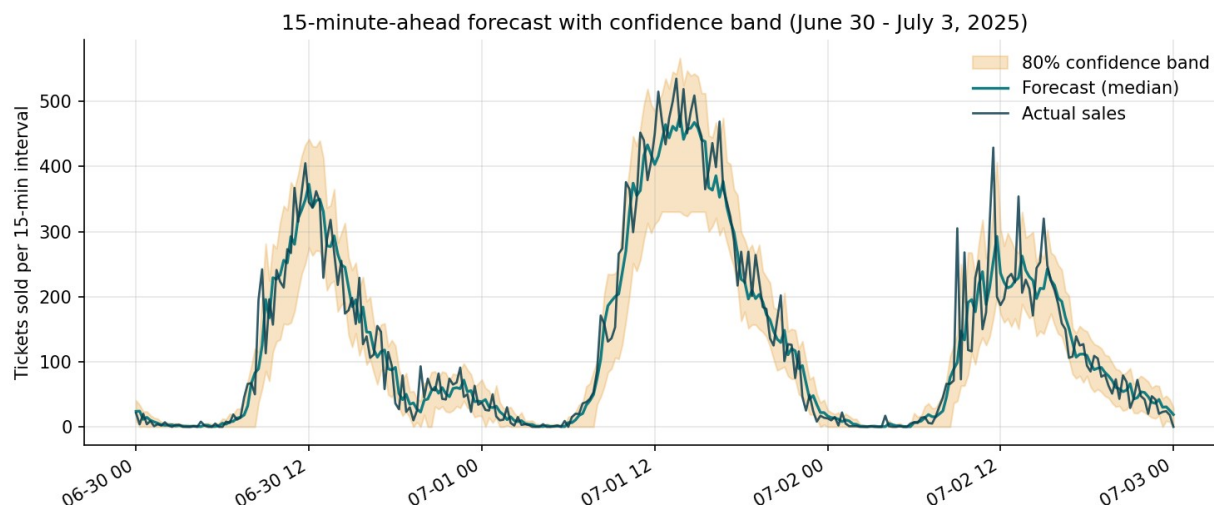


Figure 3. 15-minute-ahead forecast with 80% confidence band, June 30 - July 3, 2025 (includes Canada Day).

## 8. Discussion

Three findings stand out as most consequential for deployment. First, the advantage of machine learning over naive/statistical approaches is not fixed — it grows with forecast horizon. This is operationally significant, because the horizons of greatest planning value (1–2 hours, sufficient to reassign staff or add a sailing) are precisely where the performance gap is largest.

Second, short-horizon demand is overwhelmingly explained by very recent momentum (the trailing hour's average) rather than calendar structure. This suggests the current feature set, while effective for the horizons tested, may benefit from additional signal sources — weather, event calendars, park programming — particularly if forecasting is extended to longer horizons (e.g. next-day) in future work.

Third, forecast uncertainty in this system is highly non-stationary: a fixed-width confidence band would be inappropriate, understating risk on peak days and overstating it on quiet ones. The adaptive quantile-based approach used here more accurately reflects real operating risk.

## 9. Operational Recommendations

- Deploy Gradient Boosting with quantile-based confidence intervals as the primary forecasting engine for operational planning.
- Prioritize the 1–2 hour horizon for staffing and crowd-management decisions, given the model's comparatively stable accuracy at these horizons and the practical lead time they provide.

- Treat confidence band width itself as an operational signal — a widening band indicates rising uncertainty and may warrant closer manual monitoring, independent of the point forecast.
- Retrain the model periodically as new seasons of data accumulate, and monitor for concept drift, particularly around changes to park programming, ferry schedules, or pricing.
- Consider adding an explicit “special event” flag (e.g. Canada Day, holiday weekends) as a future feature, given observed demand anomalies tied to specific calendar dates beyond what day-of-week/month features capture.

## 10. Limitations & Future Work

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- ARIMA/SARIMA (statsmodels) and Facebook Prophet, named in the original project scope, were not evaluated due to package availability constraints in the development environment; a follow-up comparison against these approaches is recommended and outlined in the accompanying dashboard documentation.
- The deployed dashboard currently visualizes backtested historical predictions rather than a live ticketing feed; connecting a live data source is a natural next step toward production deployment.
- Redemption Count was engineered as a set of predictive features but not separately modeled as a primary forecast target; symmetric treatment of both sales and redemption forecasting is recommended for a future iteration.
- XGBoost, listed as an optional extension in the original scope, was not evaluated due to package availability; given that tree-based methods already outperformed linear approaches in this study, it represents a reasonable incremental next step.
- This analysis forecasts a single aggregate demand series; extending the approach to route-, dock-, or vessel-level granularity, if such data becomes available, would increase its direct operational applicability.

## 11. Conclusion

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This project demonstrates that short-term ferry ticket demand at Toronto Island Park is substantially predictable using machine learning methods applied to historical ticketing data alone. A Gradient Boosting model, trained on lag-based and temporal features derived from a decade of 15-minute interval data, meaningfully outperforms naive and linear baselines — and its advantage widens, rather than narrows, at the longer forecast horizons most useful for operational planning. Combined with adaptive, well-calibrated uncertainty estimates and a working interactive dashboard, these results

provide Toronto Government Parks, Forestry & Recreation with a practical foundation for shifting ferry operations from reactive to predictive planning.